SAFEMAN · STUDENT MODULE

SM-CORE

SafeMan Core

SafeMan™ by Oryx Institute · safeman.oryxinstitute.org

5-Day Flagship — The SAMTRAC-Equivalent (40 Hours) · Pass mark 70% · 8 modules

Name _____

Student no. _____ Class _____



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WELCOME

SafeMan Core

SafeMan Core — 5-Day Flagship — The SAMTRAC-Equivalent (40 Hours). This SafeMan course is engineered for corporate audit readiness against: ISO 45001:2018; ISO 14001:2015; ISO 31000; NTA Refs 846–858; GN 156 of 1997.

REGULATORY DISCLOSURE — READ FIRST

SafeMan™ is a proprietary inter-organizational safety management and operational competency benchmark developed by Oryx Institute (safeman.oryxinstitute.org). Courses are engineered in alignment with ISO 45001:2018, ISO 14001:2015, ISO 31000, ILO-OSH guidelines, and pre-referenced against Namibia Training Authority (NTA) Unit Standards and Namibian Government Notice 156 of 1997 for corporate audit readiness. SafeMan credentials verify industry competency and attendance between organizations and are non-NQA credit-bearing.

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How to use this module pack

Work through the modules in order. Each module states its focus and the key points you are expected to carry into practice. Diagrams referenced as web assets accompany the online reader at safeman.oryxinstitute.org. After the modules, sit the knowledge check: 40 questions, pass mark 70%, closed book.

MODULES

Module 1 — ISO 45001:2018 Architecture, Context & Leadership (Cl. 4–5)

The clause-by-clause skeleton of an OH&S management system and what leadership must visibly own for the system to function.

Key points

Clause 4: external/internal issues, workers and other interested parties, scope of the system.

Clause 5: leadership and worker consultation — policy, roles, responsibilities and authorities documented and visible.

The system must be integrated into business processes, not bolted on as a safety file on a shelf.

Top management demonstrates commitment with resources, protection of workers from reprisal, and participation structures.

Diagram: /assets/webp/sm-qr-shield.webp

Module 2 — ISO 31000 Risk Architecture, Baseline/Issue-Based HIRA & BowTie

Enterprise risk principles applied to OH&S: baseline, issue-based and continuous risk assessment, and barrier thinking with BowTie diagrams.

Key points

ISO 31000 process: communication, scope/context/criteria, risk assessment (identification, analysis, evaluation), treatment, monitoring, recording.

Three assessment types: baseline (whole operation), issue-based (new plant, change, incident), continuous (task and pre-use checks).

BowTie: hazard on the knot, threats left, consequences right, preventive and mitigative barriers drawn as lines of defence.

Every barrier needs an owner, a performance standard and a checking routine, or it is decoration.

Diagram: /assets/webp/sm-hira-matrix.webp

Module 3 — Safe Systems of Work, e-PTW & Contractor SHE Vetting

Converting risk assessment into rules people actually follow: safe systems of work, electronic permit-to-work and the contractor SHE file gate.

Key points

A safe system of work = task analysis, written method, competence check, supervision, and a review trigger.

PTW is required for high-hazard work: hot work, confined space, electrical isolation, excavation, working at heights.

An electronic PTW adds conflict detection: the system refuses two permits that fight each other.

Contractor SHE vetting gate: safety file, medicals, competence records, insurance, and a site induction before the first spanner turns.

Diagram: /assets/webp/sm-loto-padlock.webp

Module 4 — ISO 14001:2015 Environmental Management, Aspect/Impact & Spill Control

Environmental side of the SHEQ function: aspects and impacts registers, operational control, and first-response spill containment.

Key points

Aspect = how your activity touches the environment; impact = the change in the environment.

Significance scoring uses severity, duration, frequency and legal exposure; significant aspects get controls and objectives.

Emergency spills: contain at source, protect drains, notify per the site plan, dispose as hazardous waste with a manifest.

Waste hierarchy: avoid, reduce, reuse, recycle, treat, dispose.

Diagram: /assets/webp/sm-extinguisher.webp

Module 5 — Exposure Limits (TWA/STEL) & Hierarchy of Controls

Anticipation, recognition, evaluation and control of workplace health stressors, with the 8-hour TWA and STEL concepts done properly.

Key points

Occupational hygiene survey cycle: anticipate, recognize, evaluate, control.

TWA averages exposure over a reference shift (8 h); STEL caps short peaks (typically 15 min); ceilings are never-exceed values.

Controls in order: elimination, substitution, engineering, administrative, PPE — PPE is the last line, not the first.

Health stressors: dust (incl. silica), noise, vibration, heat, radiation, chemicals, biological agents.

Diagram: /assets/webp/sm-gas-detector.webp

Module 6 — Emergency Preparedness, Fire Engineering & Crisis Command

Building an emergency capability that survives contact with a real event: plans, equipment, command structure and drills.

Key points

Emergency plan elements: identified scenarios, alarm and communication, roles (incident controller, marshals, first aiders), assembly, roll call, handover to authorities.

Fire engineering basics: ignition source, fuel, oxidizer — remove any leg of the tetrahedron and the fire fails.

Crisis command: one voice, one board, timed decisions, a log kept from the first minute.

Drills are assessed against time objectives (e.g. full evacuation), not just performed for a photo.

Diagram: /assets/webp/sm-extinguisher.webp

Module 7 — ICAM Incident Investigation, Root-Cause Taxonomy & Cost of Loss

Investigating for system failure rather than blame: the ICAM framework, evidence handling and the real cost of loss arithmetic.

Key points

ICAM layers: organizational influences, planning and supervision, task/environment conditions, individual/team actions, and the absent or failed defences.

Evidence first: scene, positions, people, parts, papers — before memories converge and the scene is cleaned.

Root-cause taxonomy distinguishes immediate causes (energy, substance) from contributing conditions and underlying organizational factors.

Direct cost (injury, damage) is the tip; indirect cost (downtime, investigation, retraining, reputation) typically runs multiples higher.

Diagram: /assets/webp/sm-first-aid.webp

Module 8 — ISO 19011 Internal HSE Auditing, Non-Conformance Grading & Management Review

Auditing the system against its own claims: audit programme, evidence grading, findings that drive improvement, and closing the loop at management review.

Key points

Audit programme: scope, criteria, frequency by risk, auditor independence, documented method.

Evidence: objective trails (records, observations, interviews) traced against clauses — opinions are not findings.

Grade non-conformities: major (system failure or legal breach), minor (isolated lapse), observation (improvement signal).

Findings feed corrective action with owners and dates, and their trend feeds management review.

Diagram: /assets/webp/sm-qr-shield.webp

ASSESSMENT

Knowledge check

The closed-book knowledge check draws 40 questions from the bank reproduced in Appendix A. Pass mark 70%. Your SafeMan credential is issued after the knowledge check, and remains valid for 36 months from issue.

APPENDIX A — QUESTION BANK

Knowledge check bank

Q1. ISO 45001 Clause 5 requires top management to:

A. Delegate safety to the safety officer

B. Demonstrate leadership and protect workers from reprisal

C. Attend the annual audit only

D. Sign the safety file

Q2. A BowTie diagram's knot represents the:

A. Incident

B. Hazard

C. Consequence

D. Audit finding

Q3. Which risk assessment type responds to a new machine installation?

A. Baseline

B. Issue-based

C. Continuous

D. Annual review

Q4. In ISO 31000, risk treatment options include all EXCEPT:

A. Avoiding the activity

B. Reducing the risk

C. Transferring the risk

D. Hiding the risk from the register

Q5. Every BowTie barrier must have:

A. A color code

B. An owner and a performance standard

C. A laminated card

D. A monthly cost report

Q6. Continuous risk assessment is performed:

A. Once per year

B. At task level, before and during work

C. Only after incidents

D. By external consultants

Q7. An electronic permit-to-work adds most value by:

A. Sending emails

B. Detecting conflicting permits automatically

C. Replacing supervision

D. Printing faster

Q8. Before a contractor's first shift, the host must verify:

A. Their tax clearance

B. A complete SHE file, medicals and site induction

C. Their B-BBEE status

D. Their canteen menu

Q9. A safe system of work fails fastest when:

A. It is written long

B. Supervision stops enforcing it

C. It is signed

D. It is audited

Q10. 'Aspect' in ISO 14001 means:

A. The environmental change

B. How an activity interacts with the environment

C. A legal register

D. An audit clause

Q11. First response to a chemical spill is to:

A. Wash it to the drain

B. Contain at source and protect drains

C. Bury it

D. Leave until it evaporates

Q12. The waste hierarchy's first preference is:

A. Disposal

B. Treatment

C. Avoidance/reduction

D. Recycling

Q13. An 8-hour TWA is used because:

A. It shortens reports

B. Health effects track accumulated dose over a reference shift

C. Instruments need rest

D. The law copies America

Q14. STEL protects against:

A. Chronic effects

B. Short peak exposures

C. Noise only

D. Skin contact

Q15. Correct hierarchy of controls order:

A. PPE, admin, engineering, substitution, elimination

B. Elimination, substitution, engineering, administrative, PPE

C. Engineering, elimination, PPE, admin, substitution

D. Substitution, PPE, elimination, engineering, admin

Q16. Silica dust is principally dangerous because it:

A. Smells bad

B. Causes irreversible scarring of lung tissue (silicosis)

C. Stains clothes

D. Is radioactive

Q17. The fire tetrahedron's four sides are:

A. Heat, fuel, oxygen, chain reaction

B. Water, foam, CO2, powder

C. Class A to D

D. Alarm, call, fight, flee

Q18. Evacuation drills must be:

A. Announced and photographed

B. Timed against objectives and assessed

C. Cancelled in winter

D. Run by contractors

Q19. The incident controller in a crisis:

A. Reports to the fire brigade

B. Runs a single command board with a log

C. Talks to the press first

D. Leaves last with the keys

Q20. ICAM's deepest analysis layer is:

A. Individual actions

B. Task/environment conditions

C. Organizational influences

D. Absent defences

Q21. First investigation evidence to secure is:

A. The injury register

B. The scene, positions, people, parts and papers

C. The CCTV only

D. The press statement

Q22. Indirect accident costs typically:

A. Are smaller than direct costs

B. Run at multiples of direct costs

C. Are insured fully

D. Do not exist

Q23. A major non-conformity in an audit means:

A. A typo in a record

B. System failure or legal breach

C. Any late signature

D. A missing poster

Q24. An auditor must be:

A. Senior in the company

B. Independent of the area audited

C. A certified lead auditor for internal audits

D. The safety officer

Q25. Audit findings are valid when based on:

A. Feelings

B. Objective evidence traced to criteria

C. The loudest worker

D. Previous audits

Q26. Management review's verdict covers:

A. Suitability, adequacy and effectiveness

B. Salaries

C. Marketing plans

D. The canteen

Q27. Worker consultation under ISO 45001 must be:

A. Once a year at the party

B. Structured and two-way on OH&S matters

C. Only through the union

D. Anonymous only

Q28. Context of the organization (Cl. 4) examines:

A. Only the org chart

B. Internal/external issues and interested parties relevant to OH&S

C. The share price

D. The marketing mix

Q29. OH&S objectives must be:

A. Vague and inspiring

B. Measurable, resourced and monitored

C. Secret

D. Set by the client

Q30. A 'significant' environmental aspect gets:

A. A poster

B. Operational controls and possibly objectives

C. Nothing until audited

D. A fine

Q31. Elimination beats PPE because:

A. It is cheaper every time

B. It removes the hazard instead of reducing exposure

C. PPE is banned

D. Auditors prefer it

Q32. Heat stress controls include all EXCEPT:

A. Shade and hydration

B. Work-rest cycles

C. Acclimatization

D. Increasing piece-rate targets

Q33. A near-miss differs from an incident because:

A. It caused no injury or damage this time

B. It is not serious

C. It is not recorded

D. It involves no work

Q34. Barrier health is verified by:

A. Trusting the owner

B. Scheduled assurance checks against the standard

C. Annual hope

D. Posters

Q35. In contractor management, the site induction happens:

A. After the first shift

B. Before any work starts

C. When convenient

D. Only for new companies

Q36. Documented information under ISO 45001 must be:

A. Perfect and endless

B. Controlled, current and available where needed

C. Handwritten

D. Only digital

Q37. Workers must be consulted on OH&S matters because:

A. It fills minutes

B. Cl. 5.4 makes participation a system requirement

C. Clients demand it

D. It replaces training

Q38. The scope statement of an OH&S system defines:

A. The logo

B. What activities, sites and functions the system covers

C. Salaries

D. The canteen menu

Q39. Emergency preparedness under Cl. 8.2 requires:

A. A poster

B. Planned responses, training, drills and periodic testing

C. One fire extinguisher

D. A phone list

Q40. A documented information control exists mainly to:

A. Impress auditors

B. Keep records current, protected and available where needed

C. Increase paperwork

D. Store old drafts

Answer key

Q1=B Q2=B Q3=B Q4=D Q5=B Q6=B

Q7=B Q8=B Q9=B Q10=B Q11=B Q12=C

Q13=B Q14=B Q15=B Q16=B Q17=A Q18=B

Q19=B Q20=C Q21=B Q22=B Q23=B Q24=B

Q25=B Q26=A Q27=B Q28=B Q29=B Q30=B

Q31=B Q32=D Q33=A Q34=B Q35=B Q36=B

Q37=B Q38=B Q39=B Q40=B